

WASP-50b

R-1849

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_160903 · created 2026-08-03T08:04:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457634.961 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1751 +/- 0.0073
Transit depth (Rp/Rs)^2	3.07 +/- 0.26 [%]
Orbital Inclination (inc)	89.48 +/- 1.4
Airmass coefficient 1 (a1)	0.187 +/- 0.069
Airmass coefficient 2 (a2)	0.893 +/- 0.036
Scatter in the residuals of the lightcurve fit is	26.38 %
Best Comparison Star	#1 - [150, 270]
Optimal Method	PSF photometry
Transit Duration (day)	0.0245 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1751 ± 0.0073 vs 0.1404 (deviation 1.13σ)
Duration fit vs published	0.0245 d vs 0.0754 d (deviation 1.92σ)
Mid-time O–C	57.1 min
Depth SNR	5.7
Residual scatter	26.38 %

Light curve

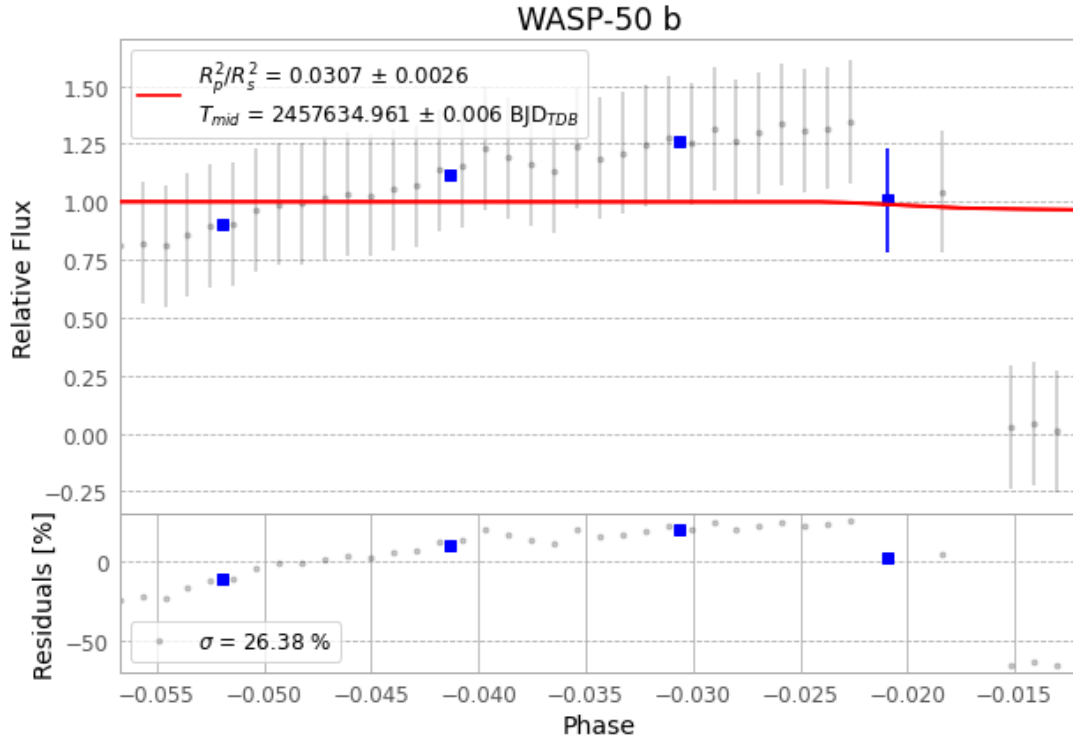


Figure 1 — FinalLightCurve_WASP-50 b_2016-09-03.png (SHA-256 3dc697bd73fa667a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [595, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 62102eeb49b26eef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-50160903081812.FITS → WASP-50160903115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-160903.log (02723819fb37...), FinalLightCurve_WASP-50 b_2016-09-03.png (3dc697bd73fa...), FinalLightCurve_WASP-50 b_2016-09-03.csv (d1b1eaca2a52...)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 08:04Z.